

Data Cleaning and Preprocessing

Internship

Codveda Technology – Data Analytics Internship

Submitted By

Name: Pawan Kushwah

College: G H Rasoni International Skill Tech University, Pune

Branch: B.Tech Artificial Intelligence & Data Science

Objective

The objective of this project is to clean and preprocess the Iris dataset by identifying missing values, removing duplicate records, checking data types, and preparing the dataset for further analysis.

Dataset

Dataset Name: Iris Dataset

The dataset contains measurements of iris flowers, including sepal length, sepal width, petal length, petal width, and species.

Tools Used

- Python
 - Google Colab
 - Pandas
 - NumPy
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Methodology

Step 1 – Import Libraries

Imported the Pandas library for data manipulation.

Step 2 – Load Dataset

Loaded the Iris dataset into a Pandas DataFrame.

Step 3 – Dataset Inspection

Used `head()`, `info()`, and `describe()` to understand the dataset structure.

Step 4 – Missing Value Detection

Checked for missing values using `isnull().sum()`.

Step 5 – Duplicate Detection

Verified duplicate records using `duplicated().sum()`.

Step 6 – Data Standardization

Reviewed categorical values for consistency.

Step 7 – Export Dataset

Saved the cleaned dataset for future analysis.

Results

- Dataset loaded successfully.
 - No missing values were found.
 - No duplicate records were found.
 - Data types were verified.
 - The cleaned dataset was saved successfully.
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Conclusion

The Iris dataset was successfully inspected and verified for data quality. Since no missing values or duplicate records were present, the dataset was already suitable for further analysis and machine learning tasks.